

# **COS30082**

## **Applied Machine Learning**



### Lecture 9

## Natural Language Processing

# What is Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a branch of artificial intelligence that focuses on the interaction between computers and humans through natural language.
- The goal of NLP is to enable machines to understand, interpret, and respond to human language. For example,
  - **Understand:** NLP systems parse customer feedback to identify key themes and sentiments.
  - **Interpret:** Determine the intent behind queries in virtual assistants.
  - **Respond:** Chatbots generate relevant responses to user inquiries in real-time.

- **Text Classification:** Identifies themes like spam or sentiment within texts.
- **Machine Translation:** Converts text or speech across languages, e.g., Google Translate.
- **Speech Recognition:** Transforms spoken words into text, used in devices like Siri or Alexa.
- **Chatbots and Virtual Assistants:** Facilitate interactive dialogues, mimicking human conversation.
- **Information Extraction:** Detects specific data like names or relationships from texts.
- **Text Summarization:** Condenses long documents into essential summaries.
- .....

# Challenges in NLP

- **Ambiguity:** Language is inherently ambiguous. For instance,
  - The word "bank" can mean a financial institution or the side of a river.
  - I can see a man with a telescope. (who has the telescope?)
- **Context Understanding:** NLP struggles with context.
  - It is a piece of cake. ( a cake or it is easy?)
- **Sarcasm and Irony:** Detecting sarcasm or irony in text is challenging.
  - Great, another rainy day! (This may be sarcastic if someone planned to go to the beach, but sincere if a farmer needs rain for crops.)

- **Language Diversity:** Different dialects and slang complicate processing. For example, "soda" vs. "pop" vs. "coke" can all refer to a soft drink, depending on the region.
- **Idioms and Phrases:** Phrases like "raining cats and dogs" (raining very heavily) are difficult to interpret literally for NLP systems.
- **Pronoun Resolution:**

“Sarah told Emily that she had lost her keys.” she refers to whom?

(Sarah lost the keys: Sarah told Emily about her own lost keys.  
Emily lost the keys: Emily did not know, and Sarah informed her.)
- **Language Variability and Evolution:** Language is constantly evolving, and new words, slang, and usage patterns emerge all the time. For example, "He is salty" means he is upset, and "He is sick" can mean he is very good at something.

# History of NLP

- 1.Early Days (1940s-1960s):** Early Models were based on rules of grammar and the formal theory of syntax.
- 2.Rise of Machine Learning (1970s-1990s):** there was a shift towards statistical methods for language processing, influenced by the limitations of rule-based systems.
- 3.The Internet Era (1990s-2000s):** The explosion of digital text data on the internet provided vast resources for training and improving NLP systems. Significant advances in machine translation during this period, driven by statistical models.

## 4. Deep Learning and LLM Era (2010s–present)

- Deep Neural Networks: CNNs, RNNs and LSTMs significantly improved NLP and outperformed earlier methods in many language tasks.
- Pre-trained Language Models: BERT and GPT used large-scale pre-training to learn rich language representations, greatly improving language understanding and text generation.
- Large Language Models: Modern LLMs are scaled-up pre-trained models, often with more parameters, larger training data, reasoning ability, and multimodal capabilities.
- Agentic AI: LLM-based systems can plan, use tools and take actions, extending NLP towards broader intelligent systems. For example:
  - Coding agents — Claude Code / GitHub Copilot: They understand coding instructions, edit files, run commands, and help complete programming tasks.
  - Smart home robot agents: They understand voice commands and can take actions in the home, such as cleaning the room, controlling lights and appliances.

# Key Terms/Concepts

**Corpus (plural: Corpora)** - A corpus refers to a large collection of text data that is used to train language models or perform other types of text analysis.

**Bag of Words (BoW)** - In the bag of words model, text is represented as a vector of word frequencies, disregarding grammar and word order. Each unique word in the text corresponds to a vector element.

# Example: Bag of Words

The Bag-of-words model is an orderless document representation where only the counts of words matter.

- I love apples and hate oranges.  
I hate bananas too.
- I love oranges and hate apples.  
I hate bananas too.

The two documents are the same in terms of words and frequency.

| Term    | Count |
|---------|-------|
| I       | 2     |
| love    | 1     |
| apples  | 1     |
| and     | 1     |
| hate    | 2     |
| oranges | 1     |
| bananas | 1     |
| too     | 1     |

# Key Terms/Concepts

**Tokenization** - The process of breaking down a text into smaller units, typically words or phrases.

**Stop Words** - Stop words are common words like "a", "the" and "is" that are often filtered out in language processing to focus on more meaningful words.

## Stemming and Lemmatisation

Both stemming and lemmatization involves reducing the inflectional forms of words to their root forms.

Inflection forms of words are words that are derived from the root or base form of a word.

For example, the words jumped, jumping and jumps are inflectional forms of the root word jump.

- **Stemming:** Stemming simplifies words to their base forms by crudely cutting off endings. It is faster but the result may not be a word.
- **Lemmatization:** Lemmatization reduces words to their dictionary form through detailed linguistic analysis, considering context and grammatical rules. It is slower but accurate as the result is a word.

# Example: Stemming and Lemmatisation

## **Word --- Stemmed word**

- jumped --- jump
- friends --- friend
- mysteries --- mysteri
- created --- creat
- took --- took

## **Word --- Lemmatized word**

- jumped --- jump
- friends --- friend
- mysteries --- mystery
- created --- create
- took --- take

**Part-of-Speech Tagging (POS Tagging)** - The process of labelling each word in a sentence with its word type, such as noun, verb, adjective, or adverb, based on its meaning and context.

For example "She can play the guitar beautifully."

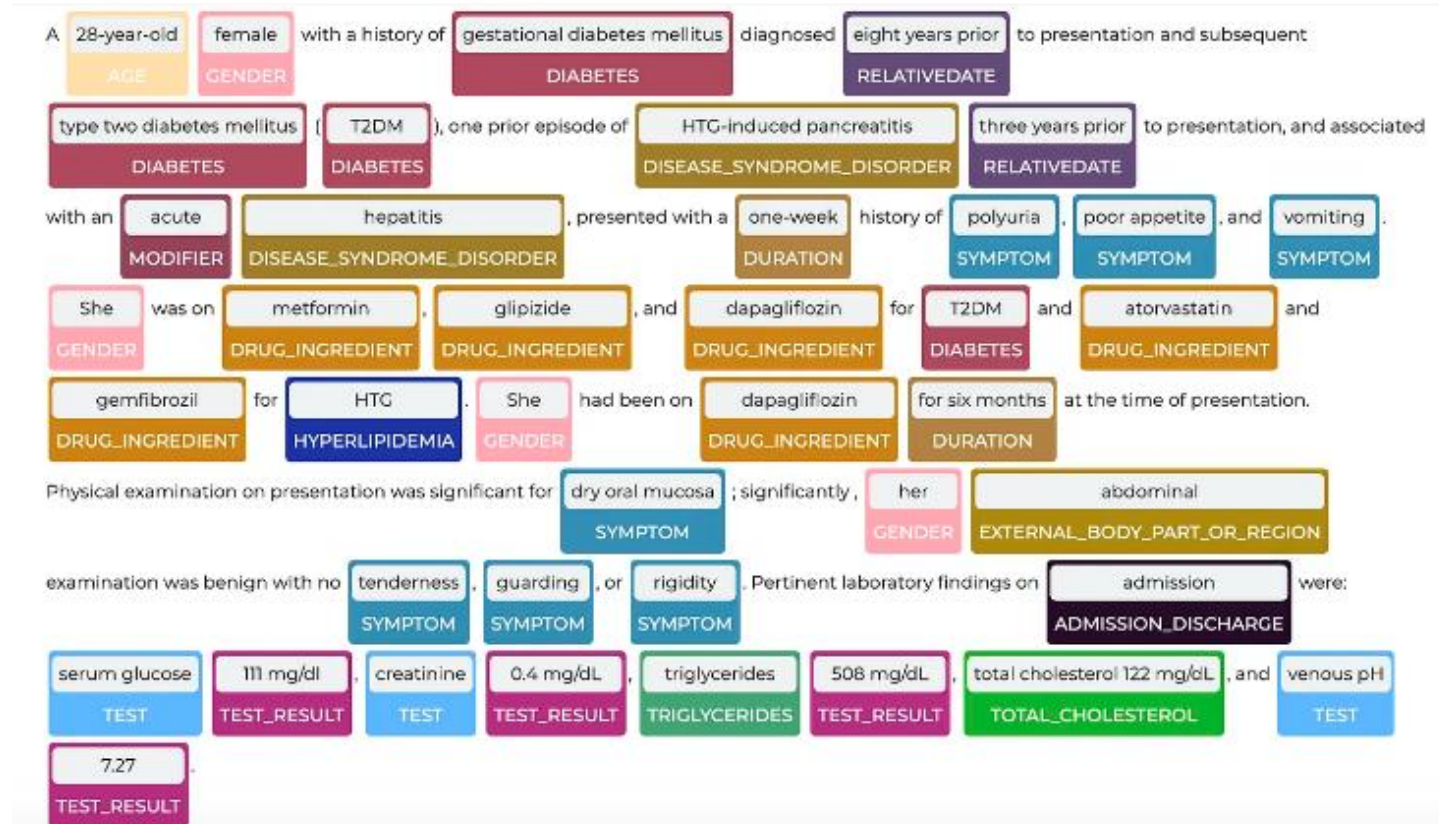
**Tagged:**

1. "She" (Pronoun)
2. "can" (Modal Verb)
3. "play" (Verb)
4. "the" (Determiner)
5. "guitar" (Noun)
6. "beautifully" (Adverb).

# Key Terms/Concepts

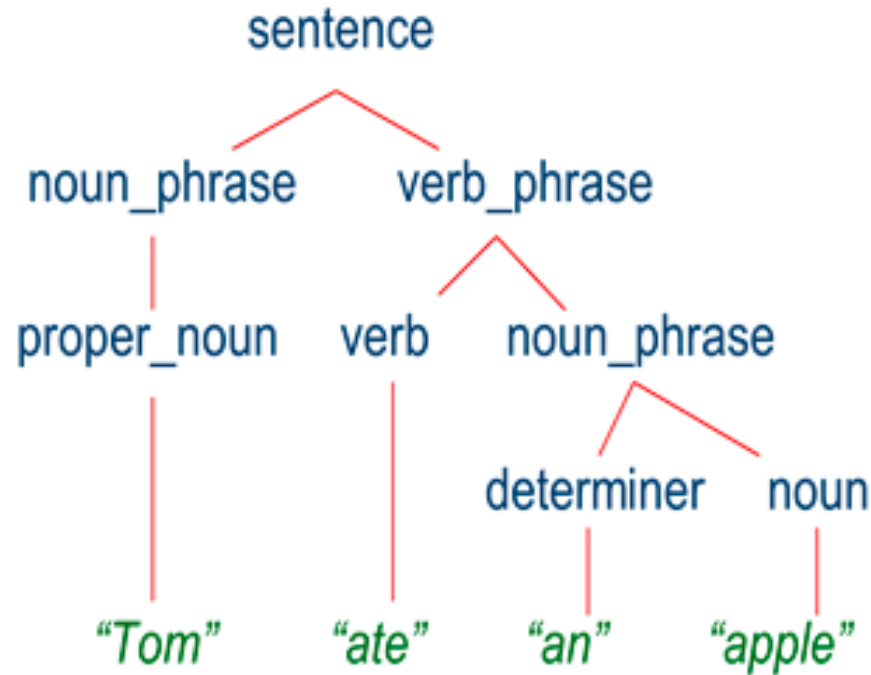
**Named Entity Recognition (NER)** - A process where an algorithm takes a string of text (sentence or paragraph) and identifies relevant nouns (people, places, and organizations) that are mentioned in that string.

Image from John Snow Lab  
Medical NER



# Key Terms/Concepts

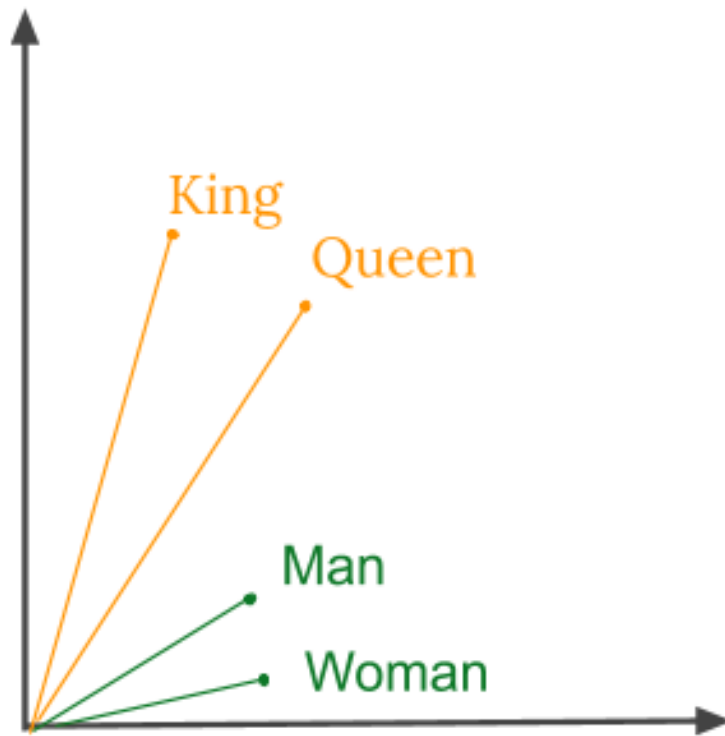
**Syntax Tree** - A tree representation of syntactic structure of sentences or strings.



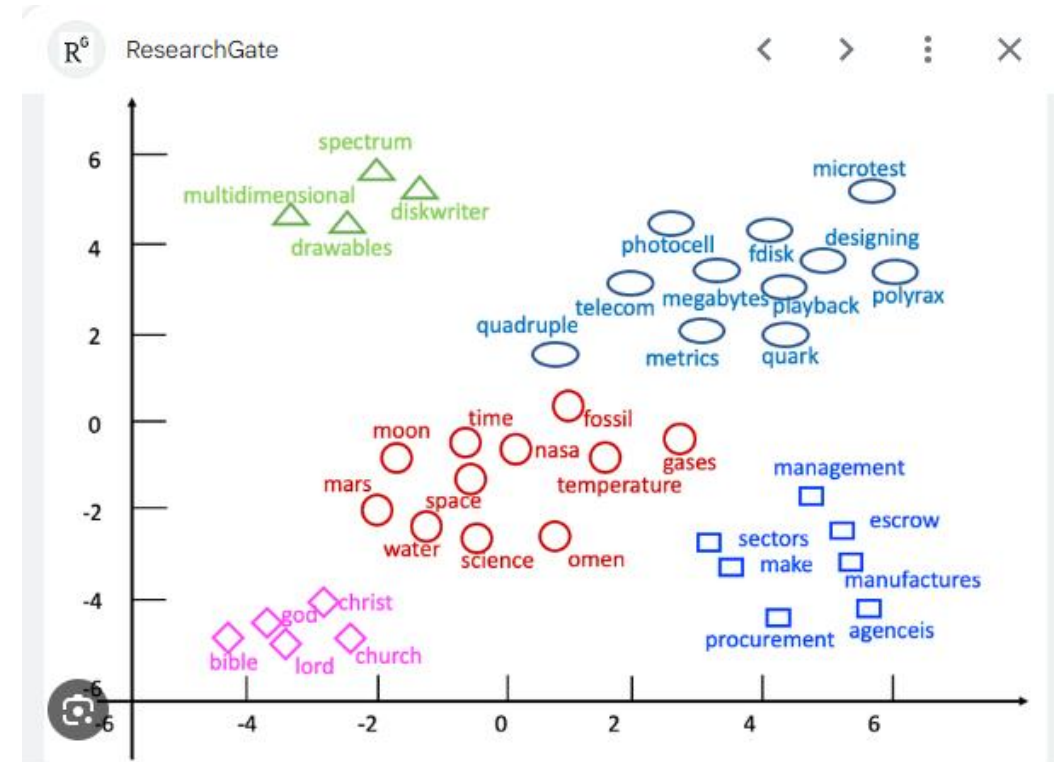
**Semantic Analysis** - An area of NLP focused on interpreting the meaning of words, sentences, and texts in context.

# Key Terms/Concepts

**Word Embeddings** - A type of word representation that allows words with similar meaning to have a similar representation.



*Similar words are closely placed in vector space*



2D PCA projection of word embeddings. Five different word clusters are... | Download Scientific Diagram

Visit >

# Key Terms/Concepts

**N-grams** - A sequence of  $n$  words, characters, or tokens in text or speech.

Example: The cat sleeps in the basket every night.

Unigram (1-gram): ["the"], ["cat"], ["sleeps"] ....

Bigram (2-gram): ["The cat"], ["cat sleeps"], ["the basket"] ...

Trigram (3-gram): ["The cat sleeps"], ["in the basket"], ...

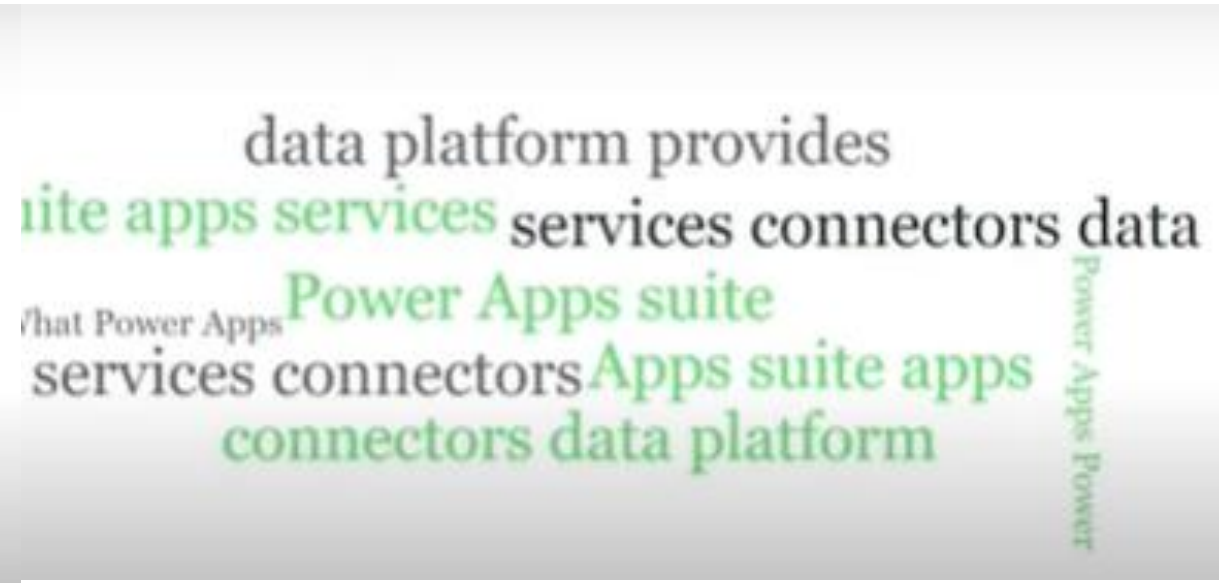
4-gram: ...

**Note:** Not all N-grams are kept. We usually filter them by frequency, keeping only sequences that appear frequently in the corpus.

By analysing frequently occurring sequences in a corpus, N-grams can capture language patterns and provide more meaningful information than single words alone.

# Key Terms/Concepts

**Word Cloud:** A word cloud visually represents text data, highlighting frequently used words in varying sizes and/or color based on their occurrence.



**TF (Term Frequency):** Measures how frequently a term occurs in a document. More occurrences increase the term's relevance in that document.

**TF-IDF (Term Frequency-Inverse Document Frequency):** Weighs a term's frequency (TF) against its rarity across all documents, enhancing its importance when it is rare.

Example: In a corpus of medical reports, the word “patient” may appear frequently in many documents, so it is not very distinctive. However, a word like “diabetes” may appear in fewer documents, but it tells us more clearly what a specific report is about.

# Key Terms/Concepts

Term Frequency (TF):

$$TF = \frac{\text{Number of times term appears in a document}}{\text{Total number of terms in the document}}$$

Inverse Document Frequency (IDF):

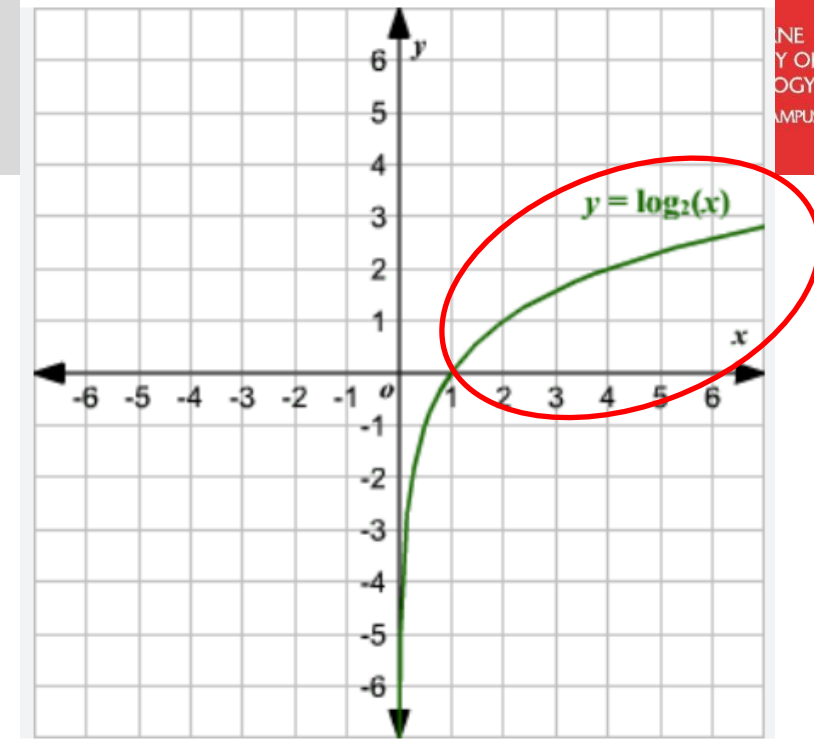
$$IDF = \log \left( \frac{\text{Total number of documents}}{\text{Number of documents with the term in it}} \right)$$

Term Frequency - Inverse Document Frequency (TF-IDF):

$$TF\text{-}IDF = TF \times IDF$$

# Key Terms/Concepts

$$\text{IDF} = \log \left( \frac{\text{Total number of documents}}{\text{Number of documents with the term in it}} \right)$$



IDF tells us how important a word is by looking at how rare it is across all documents.

If a word appears in many documents, it's common → low IDF → less important (e.g., "very", or "machine" in a ML document collection)

If a word appears in few documents, it's rare → high IDF → more important (e.g., "neuroscience", "photosynthesis").

# Example: TF-IDF

Suppose we have a collection of 100 documents, and the word “apple” appears in 5 of them.

- The IDF for “apple” would be:

$$IDF = \ln \left( \frac{100}{5} \right) = \ln(20) \approx 2.996$$

If “apple” appears 3 times in a 100-word document:

- The TF for “apple” would be:

$$TF = \frac{3}{100} = 0.03$$

Thus, the TF-IDF score for “apple” in that document is:

$$TF-IDF = 0.03 \times \ln(20)$$

$$TF-IDF \approx 0.03 \times 2.996 = 0.0899$$

If “apple” appears in 50 out of 100 documents, then:

$$IDF = \ln \left( \frac{100}{50} \right) = \ln(2) \approx 0.693$$

This means “apple” is less rare in the corpus, so its IDF weight is lower.

- The TF value (0.03) shows that “apple” has a low frequency in this document.
- However, “apple” is relatively rare in the whole corpus because it appears in only 5 out of 100 documents. Therefore, IDF gives more weight to this rare term.
- The final TF-IDF score adjusts the local frequency by the term’s rarity, increasing the importance of “apple” in this document.

# Key Terms/Concepts

**Document Term Matrix (DTM):** This is a matrix where each row represents a document and each column represents a term (word) from the corpus. The entries in the matrix typically contain the frequency of the term in the document.

**Term Document Matrix (TDM):** This matrix is the transpose of the DTM. Here, each row represents a term and each column represents a document. Similarly, the entries indicate the frequency of the term in the corresponding document.

The entries in the above matrix can include term frequency, TF-IDF, and other similar measures.

# Example: Document Term Matrix & Term Document Matrix

Document 1:

I love apples. I hate oranges.

TDM (weight is term frequency)

|        | doc 1 | doc 2 |
|--------|-------|-------|
| I      | 2/6   | 1/3   |
| Love   | 1/6   | 1/3   |
| hate   | 1/6   | 0     |
| apple  | 1/6   | 0     |
| orange | 1/6   | 1/3   |

Document 2:

I love oranges.

DTM (weight is term frequency)

|       | I   | Love | Hate | Apple | Orange |
|-------|-----|------|------|-------|--------|
| doc 1 | 2/6 | 1/6  | 1/6  | 1/6   | 1/6    |
| doc 2 | 1/3 | 1/3  | 0    | 0     | 1/3    |